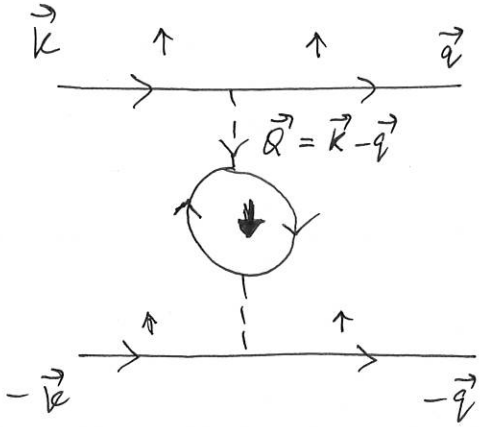
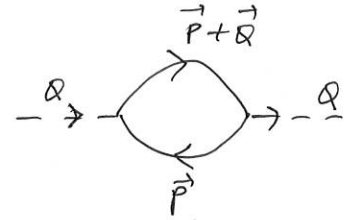
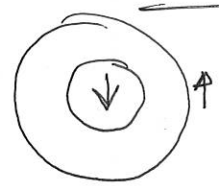
Kohn-Luttinger mechanism





Spin-polarized gas



Kagan, Chubukov JETPLett. 50 517 (1989)

Raghu, Kivelson PR B83 094518 (2011)

$$\Pi(Q) = (-) \int \frac{d\vec{p}}{(2\pi)^2} \frac{1}{\beta} \sum_{p_n} G_\downarrow(p)\, G_\downarrow(\vec{p}+\vec{Q})$$

Fermion loop

$$\frac{1}{\beta} \sum_{p_n} \int_{\vec{p}} \frac{1}{i p_n - \xi_p} \frac{1}{i p_n + i Q_n - \xi_{p+Q}} =$$

$$= \int \frac{d\vec{p}}{(2\pi)^2} \frac{1}{\beta} \sum_{p_n} \left(\frac{1}{i p_n - \xi_p} - \frac{1}{i p_n + i Q_n - \xi_{p+Q}} \right) \frac{1}{i Q_n - \xi_{p+Q} + \xi_p}$$

bosonic frequency

$$= \int \frac{d\vec{p}}{(2\pi)^2} \frac{f(\xi_p) - f(\xi_{p+Q})}{\xi_p - \xi_{p+Q} + i Q_n}$$

$$\Pi(Q_n = 0, \vec{Q}) = (-) \int \frac{d\vec{p}}{(2\pi)^2} \left(\frac{f(\xi_p)}{\xi_p - \xi_{p+Q}} - \frac{f(\xi_p)}{\xi_{p-Q} - \xi_p} \right)$$

$$\frac{\vec{p}^{\,2} - (\vec{p}+Q)^2}{2m} = \frac{-Q^2 - 2\vec{p}\cdot\vec{Q}}{2m} \qquad \frac{(\vec{p}-\vec{Q})^2 - p^2}{2m} = \frac{Q^2 - 2\vec{p}\cdot\vec{Q}}{2m}$$

$$\Pi(\vec{Q}) = (+)(2m) \int \frac{d\vec{p}}{(2\pi)^2} \left(\frac{f(\xi_p)}{Q^2 + 2\vec{p}\cdot\vec{Q}} + \frac{f(\xi_p)}{Q^2 - 2\vec{p}\cdot\vec{Q}} \right)$$

$$\cancel{\Pi = 2m \int \frac{dp\, p}{(2\pi)^2} \int d\theta\, f(\xi_p) \frac{2Q^2}{Q^4 - 2p^2Q^2\cos^2\theta}}$$

let $\vec{Q} = Q\hat{x} \Rightarrow \vec{p}\cdot\vec{Q} = Qp_x$

$$\Pi = 2m \int \frac{dp_y\, dp_x}{(2\pi)^2} \underbrace{f(\xi_p)}_{\theta(p_F^2 - p_x^2 - p_y^2)} \frac{2Q^2}{Q^4 - 4Q^2p_x^2}$$

$$\Rightarrow p_x^2 + p_y^2 \le p_F^2 \quad , \Rightarrow \int dp_y \ldots = 2\sqrt{p_F^2 - p_x^2}$$

$$\Pi = \frac{4mQ^2 \cdot 2}{(2\pi)^2} \int_{-p_F}^{p_F} dp_x \frac{\sqrt{p_F^2 - p_x^2}}{Q^4 - 4Q^2p_x^2} = (p_x = p_F x)$$

$$= \frac{8mQ^2\, p_F^2}{4\pi^2 \cdot 4Q^2p_F^2} \cdot 2\int_0^1 dx \frac{\sqrt{1-x^2}}{\frac{Q^2}{4p_F^2} - x^2} = \frac{m}{\pi^2}\int_0^1 dx \frac{\sqrt{1-x^2}}{a^2 - x^2}$$

$$a = \frac{Q}{2p_F}$$

let $y = \sqrt{1-x^2}$, $\cancel{dy = \frac{-2x\,dx}{2\sqrt{1-x^2}} \Rightarrow dx = -dy\frac{\sqrt{1-x^2}}{x} =}$

$y^2 = 1 - x^2$, $\cancel{x = \sqrt{1-y^2}}$ $\cancel{= -\frac{y}{\sqrt{1-y^2}}dy}$

$$\cancel{\Pi = \frac{m}{\pi^2}\left\{-\int_1^0 \frac{dy\, y}{\sqrt{1-y^2}} \frac{y}{a^2 - 1 + y^2}\right\} =}$$

$$\cancel{= \frac{m}{\pi^2} \int}$$

$x = \sin\theta \quad , dx = \cos\theta \, d\theta$

$$\Pi = \frac{m}{\pi^2} \int_0^{\pi/2} d\theta \, \frac{\cos^2\theta}{a^2 - \underbrace{\sin^2\theta}_{1-\cos^2\theta}} = \left(\frac{m}{\pi^2}\right) \int_0^{\pi/2} d\theta \, \frac{\cos^2\theta}{\underbrace{a^2-1}_{\beta} + \cos^2\theta} =$$

$$= \left(\frac{m}{\pi^2}\right) \int_0^{\pi/2} d\theta \, \frac{\beta + \cos^2\theta - \beta}{\beta + \cos^2\theta} = \frac{m}{\pi^2}\left(\frac{\pi}{2} - \beta \, J\right)$$

$$J = \int_0^{\pi/2} d\theta \, \frac{1}{\beta + \cos^2\theta} = \int_0^{\pi/2} d\theta \, \frac{2}{2\beta + 1 + \cos 2\theta} =$$

$$\hookrightarrow \frac{1}{2}(\cos 2\theta + 1)$$

$$= \int_0^{\pi} \frac{d\varphi}{2\beta + 1 + \cos\varphi} = \frac{1}{2\beta+1} \int_0^{\pi} \frac{d\varphi}{1 + \alpha\cos\varphi} =$$

$$\alpha \equiv \frac{1}{2\beta+1}$$

$$= \frac{1}{2\beta+1} \, \frac{\pi \, \theta(1-\alpha^2)}{\sqrt{1-\alpha^2}} =$$

(square root is REAL)

$$= \frac{\pi}{\sqrt{(2\beta+1)^2 - 1}}$$

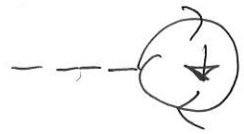
$$\Pi = \frac{m}{\pi^2}\left(\frac{\pi}{2} - \frac{\pi\beta}{\sqrt{4\beta^2 + 4\beta}}\right) = \frac{m}{2\pi}\left(1 - \sqrt{\frac{a^2-1}{a^2}}\right)$$

$\beta = a^2 - 1$

$\beta(1+\beta) = (a^2-1)a^2$

$$= \frac{m}{2\pi}\left(1 - \sqrt{1 - \left(\frac{2p_F}{Q}\right)^2}\right).$$

$$= \frac{m}{2\pi}\left[1 - \frac{\sqrt{Q^2 - 4p_F^2}}{Q}\right]$$

$\Rightarrow$  describes effective interaction between $\uparrow$ electrons, mediated by $\downarrow$ ones.

$$V_{\uparrow\uparrow}(\vec{k},\vec{q}) = -\Pi_{\downarrow\downarrow}(\vec{Q}=\vec{k}-\vec{q})\cdot u^2$$

$$\vec{Q}=\vec{k}-\vec{q} \Rightarrow Q=\sqrt{k^2+q^2-2kq\cos\varphi} = \sqrt{2}\,p_\uparrow\sqrt{1-\cos\varphi} =$$

$$= 2p_\uparrow\left|\sin\frac{\varphi}{2}\right| =$$

$$= 2p_\uparrow \sin\frac{\varphi}{2}$$

since $\varphi\in(0,2\pi)$

$k, q = p_\uparrow$

$1-\cos\varphi = 2\sin^2\frac{\varphi}{2}$

However $\Pi_{\downarrow\downarrow} = +\nu\left(1-\sqrt{1-\frac{4p_\downarrow^2}{Q^2}}\right)$

where $p_\downarrow$ = Fermi momentum of $\sigma=\downarrow$ electrons!

$$p_\downarrow < p_\uparrow .$$

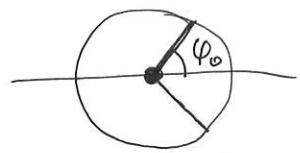
$$V_{\uparrow\uparrow} = -\nu u^2\left\{1-\sqrt{1-\frac{4p_\downarrow^2}{4p_\uparrow^2\sin^2\frac{\varphi}{2}}}\right\} =$$

$$= -\nu u^2 + \nu u^2\,\frac{\sqrt{\sin^2\frac{\varphi}{2}-\eta^2}}{\sin\frac{\varphi}{2}} \quad ; \quad \eta \equiv \frac{p_\downarrow}{p_\uparrow} \leq 1$$

Angular harmonics $(m = 1, 3, 5 \ldots)$

$$\lambda_{m\uparrow} = \int_0^{2\pi} \frac{d\varphi}{2\pi} V_{\uparrow\uparrow}(\varphi) \cos(m\varphi) = \frac{\nu u^2}{2\pi} \cdot 2 \int_{\varphi_0}^{\pi} d\varphi \cos(m\varphi) \cdot$$

$$\cdot \frac{\sqrt{\sin^2\frac{\varphi}{2} - \eta^2}}{\sin\frac{\varphi}{2}}$$



where $\sin\frac{\varphi_0}{2} = \eta$, $\varphi_0 = 2\sin^{-1}(\eta) \leq \pi$.

<u>let m=1</u> $$\lambda_1 = \frac{\nu u^2}{\pi} \int_{\varphi_0}^{\pi} d\varphi \frac{\left(1 - 2\sin^2\frac{\varphi}{2}\right)}{\sin\frac{\varphi}{2}} \sqrt{\sin^2\frac{\varphi}{2} - \eta^2}$$

$x = \cos\frac{\varphi}{2}$, $dx = -\frac{1}{2}\sin\frac{\varphi}{2}\, d\varphi$

$\cancel{dx = -\frac{1}{2}\cos\frac{\varphi}{2}\, d\varphi}$

$$-2\int_{\sqrt{1-\eta^2}}^{0} d\left(\frac{\varphi}{2}\right) \frac{\sin\frac{\varphi}{2}}{\sin^2\frac{\varphi}{2}} \left(2\cos^2\frac{\varphi}{2} - 1\right) \sqrt{1 - \cos^2\frac{\varphi}{2} - \eta^2} =$$

$$= -2 \int \frac{d\left(\cos\frac{\varphi}{2}\right)}{1 - \cos^2\frac{\varphi}{2}} \left(2\cos^2\frac{\varphi}{2} - 1\right) \left(1 - \eta^2 - \cos^2\frac{\varphi}{2}\right)^{1/2} =$$

$$= -2 \int_{\sqrt{1-\eta^2}}^{0} dy \frac{2y^2 - 1}{1 - y^2} \sqrt{1 - \eta^2 - y^2} =$$

$$= \left\{ 2\int dy \frac{2(y^2-1)+1}{1-y^2} \sqrt{1-\eta^2-y^2} = -4\int dy \sqrt{1-\eta^2-y^2} \right.$$

$$\left. + 2\int dy \frac{\sqrt{1-\eta^2-y^2}}{1-y^2} \right\} \equiv -I$$

$$\sin\frac{\varphi_0}{2} = \eta \Rightarrow \cos\frac{\varphi_0}{2} = \sqrt{1-\eta^2}\ ; \quad \int_{\sqrt{1-\eta^2}}^{0} dy \xrightarrow[y=\sqrt{1-\eta^2}\cos\theta]{} \int_{0}^{\pi/2} d\theta$$

$$y = \sqrt{1-\eta^2}\cos\theta \Rightarrow (1-\eta^2)^{1/2} \longmapsto \theta = 0\ ;\ 0 \longmapsto \theta = \pi/2$$

$$I_1 = -4\int_0^{\pi/2} \sqrt{1-\eta^2}\,\left(-\sin\theta\, d\theta\right)\,(1-\eta^2)^{1/2}\sqrt{1-\cos^2\theta} =$$

$$= +4\,(1-\eta^2)\int d\theta\, \sin^2\theta = 4(1-\eta^2)\left(\frac{\theta}{2} - \frac{1}{4}\sin 2\theta\right)$$

$$I_2 = 2\,(1-\eta^2)^{1/2}\int -\sin\theta\, d\theta\, \frac{\sqrt{1-\eta^2}\sqrt{1-\cos^2\theta}}{1-(1-\eta^2)\cos^2\theta} =$$

$$= -2\,(1-\eta^2)\int d\theta\, \frac{\sin^2\theta}{1-\cos^2\theta+\eta^2\cos^2\theta}$$

$$\int d\theta\, \frac{\sin^2\theta}{\sin^2\theta + \eta^2(1-\sin^2\theta)} = \int d\theta\, \frac{\sin^2\theta}{(1-\eta^2)\sin^2\theta+\eta^2}$$

$$= \int d\theta\, \frac{(1-\eta^2)\sin^2\theta + \eta^2 - \eta^2}{(1-\eta^2)\left[(1-\eta^2)\sin^2\theta+\eta^2\right]} =$$

$$= \frac{1}{1-\eta^2}\int d\theta \left\{1 - \frac{\eta^2}{\eta^2\left(\frac{1-\eta^2}{\eta^2}\sin^2\theta+1\right)}\right\}$$

$$= \frac{\theta}{1-\eta^2} - \frac{1}{1-\eta^2}\int \frac{d\theta}{1+\frac{1-\eta^2}{\eta^2}\sin^2\theta}$$

But let now $\frac{1-\eta^2}{\eta^2} = a^2$, $1+a^2 = 1 + \frac{1-\eta^2}{\eta^2} = \frac{1}{\eta^2}$

$$\int \frac{d\theta}{1+a^2\sin^2\theta} = \int \frac{d\theta}{(1+a^2)\sin^2\theta - \sin^2\theta + 1} =$$

$$= \int \frac{d\theta}{\cos^2\theta + (1+a^2)\sin^2\theta} = \int \underbrace{\left(\frac{d\theta}{\cos^2\theta}\right)}_{= d(\tan\theta)} \frac{1}{1+(1+a^2)\tan^2\theta} =$$

$$= \int \frac{d(\tan\theta)}{1+(1+a^2)\tan^2\theta} = \int \frac{du}{1+(1+a^2)u^2} =$$

$$= \left(u = \frac{y}{\sqrt{1+a^2}}\right) = \frac{1}{\sqrt{1+a^2}} \int \frac{dy}{1+y^2} = \frac{\tan^{-1}(y)}{\sqrt{1+a^2}}$$

$$= \frac{\tan^{-1}\left(\sqrt{1+a^2}\,\tan\theta\right)}{\sqrt{1+a^2}} = \eta \tan^{-1}\left(\frac{\tan\theta}{\eta}\right).$$

$$I_1 = (1-\eta^2)\,(2\theta - \sin 2\theta)$$

$$I_2 = -2\,(1-\eta^2)\left(\frac{\theta}{1-\eta^2} - \frac{1}{1-\eta^2}\,\eta\,\mathrm{Tan}^{-1}\left(\frac{\tan\theta}{\eta}\right)\right)$$

$$I_1\Big|_0^{\pi/2} = (1-\eta^2)\,(\pi)$$

$$I_2\Big|_0^{\pi/2} = (-2)\left[\theta - \eta\,\mathrm{Tan}^{-1}\left(\frac{\tan\theta}{\eta}\right)\right]\Bigg|_0^{\pi/2} =$$

$$= (-2)\left[\frac{\pi}{2} - \eta\,\mathrm{Tan}^{-1}\left(\frac{\tan(\frac{\pi}{2})}{\eta}\right)\right] =$$

$$= (-2)\left(\frac{\pi}{2} - \eta\,\frac{\pi}{2}\right) = \pi\,(\eta - 1) = -\pi\,(1-\eta)$$

$$I = I_1 + I_2 = \pi\,\left(1-\eta^2 - (1-\eta)\right) = \pi\,(1-\eta)\left[1+\eta-1\right] =$$

$$= \pi\,\eta\,(1-\eta)$$

$$\Rightarrow \lambda_{1\uparrow} = \frac{\nu U^2}{\pi}\,(-I) = -\nu U^2\,\eta\,(1-\eta).$$

$$= -\nu U^2\left[\frac{1}{4} - \left(\eta - \frac{1}{2}\right)^2\right]$$

$$\eta^2 - \eta = \left(\eta - \frac{1}{2}\right)^2 - \frac{1}{4}$$

$$\Downarrow$$

max at $\eta = \frac{1}{2}$.

$\Rightarrow$ superconducting critical temperature for $\sigma = \uparrow$ p-wave SC

$$T_{c} \sim E_F\, e^{-1/|\lambda_{1\uparrow}|}.$$